

FORM TP 2008143



TEST CODE 22201020

MAY/JUNE 2008

**CARIBBEAN EXAMINATIONS COUNCIL
ADVANCED PROFICIENCY EXAMINATION**

ACCOUNTING

UNIT 2 – PAPER 02

$2\frac{3}{4}$ hours

08 JULY 2008 (p.m.)

1. This paper consists of **THREE** questions.
2. **EACH** question is worth 35 marks.
3. **ALL** questions are **COMPULSORY**.
4. Begin **EACH** answer on a new page.
5. Silent non-programmable calculators may be used, but **ALL** necessary working should be clearly shown.

1. Cox Manufacturing Company produces seat belts for automobile companies. During the year just ended, the company produced 60 000 seatbelts and incurred the following costs :

Materials and supplies used	\$
Strapping	360 000
Metal clasps	127 500
Thread	4 500
Machine lubricants	1 500
Wages and salaries	
Machine operators	333 000
Production supervisors	100 500
Other factory costs	
Maintenance personnel	20 000
Janitorial personnel	24 000
Insurance	33 000
Administrative expenses (50% fixed)	165 000
Selling expenses (100% fixed)	148 500

- (a) Compute EACH of the following:

- | | | |
|-------|-----------------------------------|------------|
| (i) | Total direct materials | [2 marks] |
| (ii) | Total indirect labor | [2 marks] |
| (iii) | Total factory overhead | [3 marks] |
| (iv) | Total product cost | [2 marks] |
| (v) | Total period cost | [2 marks] |
| (vi) | Total variable manufacturing cost | [2 marks] |

- (b) (i) Cox Manufacturing Company recently received an order for 10 000 seatbelts from an automobile manufacturer in Anguilla.

Assuming that Cox Manufacturing Company has the production capacity to fill this order without affecting existing production, **list the categories of costs** that are relevant to the decision. [2 marks]

- (ii) Cox Manufacturing will not accept the order if the company will be worse off financially by so doing. Calculate the minimum price that Cox Manufacturing Company can charge for the order. [3 marks]

GO ON TO THE NEXT PAGE

- (c) Cox Manufacturing Company has three service departments and two operating departments. Selected data for 2007 is presented below.

	Service Departments			Operating Departments		Total
	Administration	Janitorial	Maintenance	Cutting & Stitching	Assembly	
Overhead costs	\$140 000	\$105 000	\$48 000	\$275 000	\$430 000	\$998 00
Number of employees	60	35	140	315	210	760
Square feet of space occupied	15 000	10 000	20 000	40 000	100 000	185 000
Hours of machine time	—	—	—	60 000	30 000	90 000

The company allocates service department costs as follows:

Administration – based on number of employees

Janitorial – based on space occupied

Maintenance – based on hours of machine time.

Cox Manufacturing Company uses the direct method to allocate service department costs.

- (i) Calculate the allocation rate for EACH service department. [3 marks]
- (ii) Calculate the amount of overhead cost that will be assigned to EACH operating department. [14 marks]

Total 35 marks

2. (a) Williams and Associates, is a prestigious law firm owned by Donna Gray and Indira Rampersad in St. Georges, Grenada. The firm uses a job order costing system to monitor the cost of servicing its clientele. The office manager has prepared the following budget for 2007:

Client billings		\$12 672 000
Less: Professional staff costs (85%)	\$6 600 000	
Administrative staff costs (75%)	2 200 000	
Other office costs (60%)	<u>1 100 000</u>	<u>9 900 000</u>
Operating income		<u>\$ 2 772 000</u>

The numbers in brackets indicate the percentage of cost that can be directly traced (traceable costs) to client jobs. The remaining portion of each cost that cannot be directly traced to specific jobs (non-traceable costs) is charged to client jobs by using a predetermined overhead application rate. The office manager uses total direct costs as the overhead application base.

In March, the firm completed work on a lawsuit for Eldon Company. The following costs were directly traced to Eldon Company:

Professional staff	\$25 000
Administrative staff	6 400
Other office costs	6 600

By the end of 2007, the actual costs directly traced to client jobs amounted to \$8 305 000 and non-traceable costs incurred totaled \$2 139 500.

- (i) Calculate the firm's pre-determined overhead application rate. [4 marks]
- (ii) Prepare a job cost sheet for the Eldon Company job showing clearly the **traceable** and **non-traceable** costs for the job. [7 marks]
- (iii) Calculate the amount of over- or under-applied overhead (non-traceable) costs for Williams and Associates for 2007. [5 marks]

- (b) Eldon Company, one of the clients of Williams and Associates, produces and sells a single product, a hand press for moulding pastries. Selected cost and operating data relating to the product for one year are provided below:

Selling price per unit		\$ 50
Manufacturing costs:		
	Variable per unit produced:	
	Direct materials	11
	Direct labour	6
	Variable overhead	3
	Fixed per year	120 000
Selling and administrative costs:		
	Variable per unit sold	5
	Fixed per year	70 000

	Units
Units in beginning inventory	0
Units produced during the year	10 000
Units sold during the year	8 000
Units in ending inventory	2 000

- (i) Compute the unit product cost assuming that Eldon Company uses
 - a) absorption costing [3 marks]
 - b) variable costing [2 marks]
 - (ii) Prepare a variable costing income statement for the company. [5 marks]
 - (iii) Prepare an absorption costing income statement for the company. [5 marks]
 - (iv) Reconcile the difference between the net operating income computed under variable costing and that computed under absorption costing. [2 marks]
- (c) Both Eldon Company and Williams and Associates use traditional costing methods. Outline ONE way in which activity-based costing differs from traditional costing methods. [2 marks]

Total 35 marks

3. (a) (i) Describe FOUR benefits that a business may gain from budgeting. [4 marks]
- (ii) Distinguish between 'ideal' and 'practical standards'. [4 marks]
- (iii) Identify the factors that should be considered in setting the standard cost for a product or service. [3 marks]
- (b) Pamela Codrington plans to open a day care centre for young children at the beginning of the new year. She rented a building and has estimated that the following annual costs will be required to operate the centre:

Salaries for three teachers	\$55 200
Salaries for two assistants	\$19 200
Rent	\$19 200
Utilities	\$ 4 320
Miscellaneous	\$ 2 880
Supplies	\$48 per child
Meals and snacks	\$456 per child

Based on the space in the building and the number of teachers, Pamela believes that the maximum number of children that can be cared for is 180. Pamela plans to charge a fee of \$1 400 per child.

- (i) Compute the contribution margin. [2 marks]
- (ii) Calculate the number of children who must be enrolled in order for the centre to break even. [4 marks]
- (iii) Calculate the margin of safety in **dollars AND** as a **percentage** of budgeted revenue, if the budgeted enrolment of the day care centre is 135 children. [5 marks]

- (c) Cork Hill Company produces three products, BP, PM, and WD. Selected data relating to the three products are presented below:

	Products (per unit)		
	BP	PM	WD
Selling price	\$80	\$56	\$70
Less: Variable expenses			
Direct materials	24	15	9
Other variable expenses	24	27	40
Total variable expenses	48	42	49
Contribution margin	\$32	\$14	\$21
Contribution margin ratio	40%	25%	30%
Estimate demand during the following month	500 units	600 units	1000 units

The same material is used in each product. The material costs \$3 per kilogram (kg) with a maximum of 5,000 kilograms (kgs) available each month.

- (i) Determine the amount of raw materials (in kg) needed to produce ONE unit of EACH product. [3 marks]
- (ii) Determine the contribution margin per kg of material used in EACH product. [3 marks]
- (iii) Prepare a schedule showing the **total** direct materials that will be required to produce the units estimated to be sold during the following month. [3 marks]
- (iv) Cork Hill Company would like to maximize its income from the production of the three products. State the criteria that should be used by the company to allocate the 5 000 kg of raw materials among the three products. [1 mark]
- (v) Determine the amount of EACH product that Cork Hill Company should produce during the following month. [3 marks]

Total 35 marks

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